



Department of Information Technology

Academic Year 2025–2026 (Odd Semester)

Degree, Semester & Branch : V Semester, B. Tech, IT
Course Code / Title : CS3551/Distributed Computing
Name of the Faculty member : Mrs. A. Alagulakshmi, AP/IT

Innovative Practice Description

- **Unit/ Topic:** Unit V - Virtualization
- **Course Outcome:** CO5
- **Activity Chosen:** Flipped Classroom
- **Justification:**
The flipped classroom approach for the topic “Types of Virtualization and Hypervisors” promotes active and personalized learning by enabling students to explore foundational concepts—such as different types of virtualizations (system, storage, network, and application virtualization), the role of hypervisors (Type 1 and Type 2), and their architecture—prior to the class through a PPT shared in Canvas earlier. This independent exploration allows students to understand how virtualization enables efficient resource utilization, scalability, and isolation in distributed systems. Overall, the flipped classroom cultivates a deeper and more interactive learning experience for understanding virtualization concepts. Peer-to-peer discussions lead to richer insights and improved conceptual clarity.
- **Time Allotted for the Activity:** 10Minutes
- **Details of the Implementation:**
 - Relevant learning resources such as PPT was uploaded to Canvas two days in advance to allow students early access.
 - Students were instructed to review the content independently and note down key points based on their understanding.
 - During the in-class session, learners were grouped into teams of four and given 10 minutes for collaborative discussion on the topic.
 - Following that, a representative from each team presented their collective insights to the whole class, as depicted in Figures 1 and 2.

- **CO–PO/ PSO mapping:**

CO	PO1	PO2	PO3	PO4	PO5	PO10	PO12	PSO1	PSO2	PSO3
CO5	3	1	1	1	2	2	2	3	3	2

(1– Low 2 –Moderate 3 – High)

- **PO/ PSO mapped:**

Innovative practice	PO1	PO2	PO3	PO4	PO5	PO10	PO12	PSO1	PSO2	PSO3
CO5	3	1	1	1	2	2	2	3	3	2
Justification for correlation	Requires fundamental engineering knowledge to understand system architecture and virtualization concepts such as hardware-level and system-level virtualization .	Involves analysing how virtualization can be applied to solve complex computing problems efficiently.	Helps students design virtualized environments to optimize resource utilization and scalability in distributed systems.	Encourages students to investigate and compare different hypervisor architectures and performance metrics.	Promotes the use of modern virtualization tools (e.g., VMware, VirtualBox) for hands-on learning and implementation.	Enhances communication skills through team discussions and presentations in the flipped classroom activity.	Encourages life-long learning as virtualization is a rapidly evolving technology in cloud and distributed computing.	Enables students to apply computing knowledge in designing and implementing virtualized IT infrastructures.	Develops skills to utilize emerging technologies like cloud-based virtualization platforms for IT solutions.	Helps students understand and apply virtualization concepts in distributed and cloud computing environments.

- **Images/ Screenshot of the practice:**

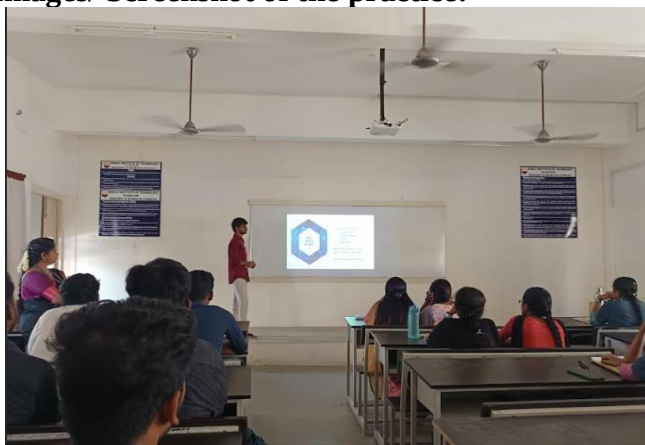


Figure 1



Figure 2

Figure 1 and 2 shows the Glimpses of presentation done by Vijayakumar R during the flipped class room Activity.

- **Reflective Critique:**

- ❖ ***Feedback of practice from students and other stakeholders:***

- Students reported a better understanding of types of virtualizations and the role of hypervisors after peer discussions.
- Many students expressed that the flipped classroom helped them visualize how virtualization enables efficient resource utilization in distributed systems.
- Students appreciated that reviewing the PPT beforehand allowed them to engage more confidently during in-class discussions.

- ❖ ***Benefit of the practice:***

- Encourages active participation and critical thinking in understanding system and hardware-level virtualization.
- Enhances conceptual clarity through peer explanation and group presentations.
- Promotes collaboration and communication among students during group activities.
- Enables personalized teacher support by addressing students' specific doubts during discussion time.
- Develops independent learning skills and improves technical vocabulary related to virtualization and distributed computing.

- ❖ ***Challenges faced in implementation:***

- Some students found it hard to understand technical content without direct instruction.
- Not all students contributed equally in group tasks, affecting collaboration.

References:

- ❖ <https://www.teachthought.com/learning/the-definition-of-the-flipped-classroom/#:~:text=A%20flipped%20classroom%20is%20a,the%20students%20independently%20at%20home.>
- ❖ https://en.wikipedia.org/wiki/Flipped_classroom

Signature of Faculty Member

HOD